KAIZEN MATHS

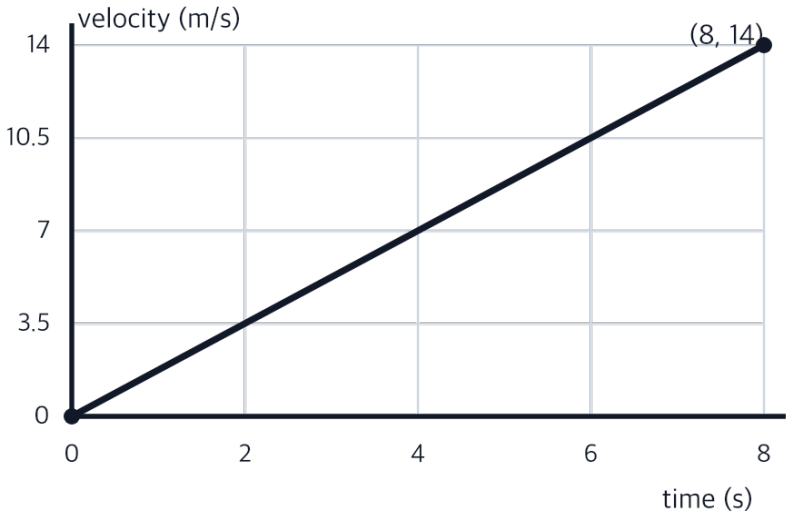
# Lesson starter

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name _____

**SA1**




The velocity-time graph shows an object accelerating from rest. Find the distance travelled.

MEC-50A8-L2 · SA

Show your working

KAIZEN MATHS

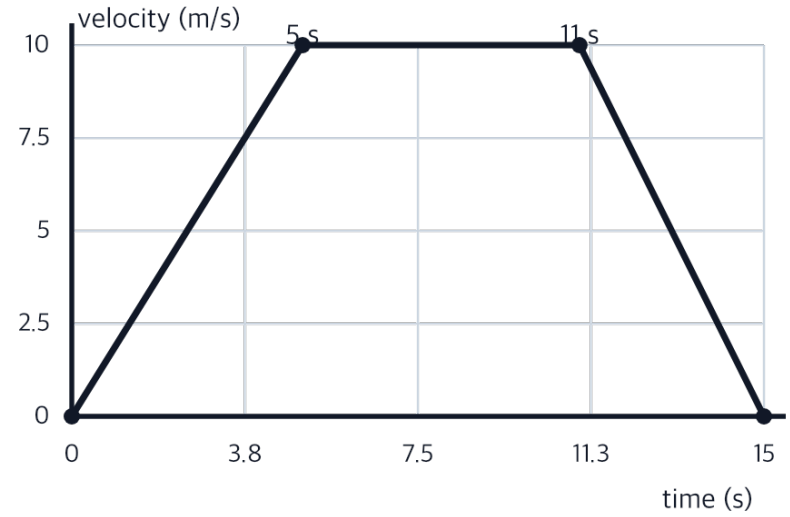
# Lesson starter

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name _____

**SB1**




Use the velocity-time graph to find the total distance travelled and the average speed for the whole journey.

MEC-50A8-L2 · SB

Show your working

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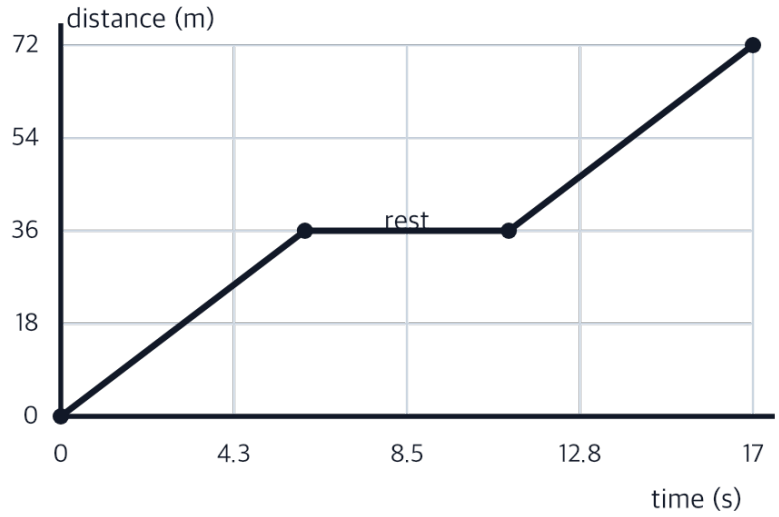
# Lesson starter

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name _____

SC1




Use the distance-time graph to state when the object is stationary and find the speed on the final section.

MEC-50A8-L2 · SC

Show your working

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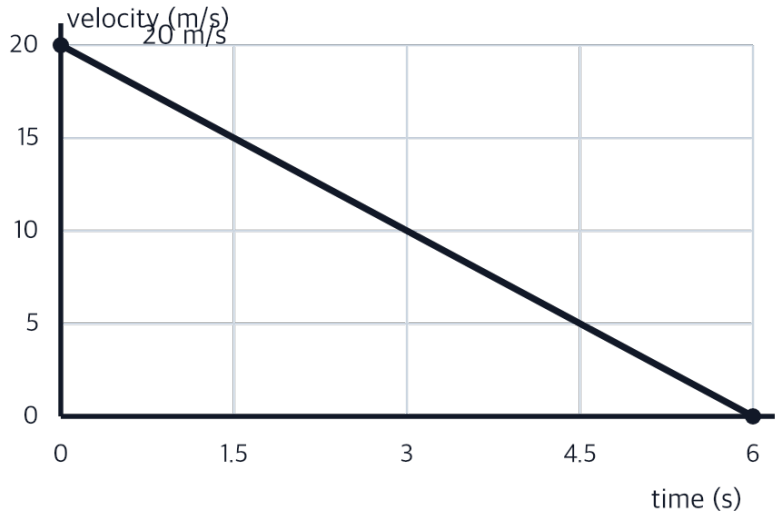
Lesson starter

Motion Graphs and Constant Acceleration

Motion Graphs

Name _____

SD1




Use the velocity-time graph to find the acceleration. State the deceleration as a positive value.

MEC-50A8-L2 · SD

Show your working

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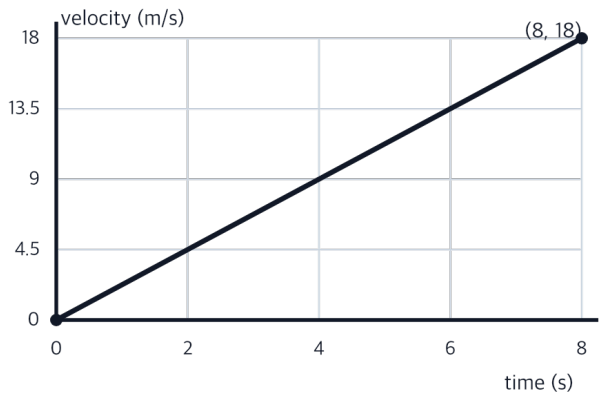
# Exit ticket

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name

**EA1**




The velocity-time graph shows an object accelerating from rest. Find the distance travelled.



Show your working

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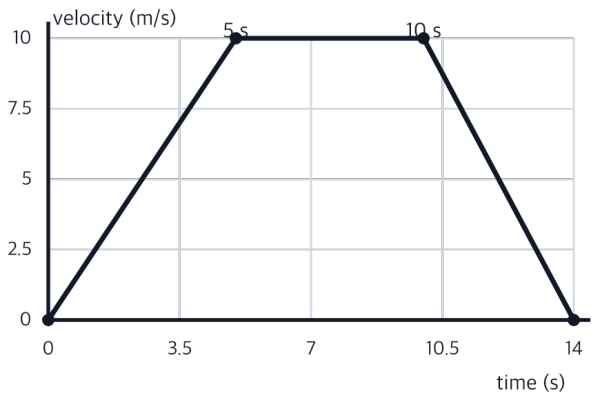
# Exit ticket

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name

**EB1**




Use the velocity-time graph to find the total distance travelled and the average speed for the whole journey.



Show your working

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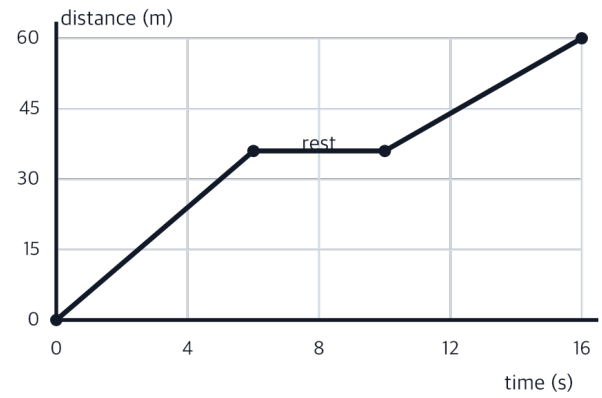
# Exit ticket

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name

**EC1**




Use the distance-time graph to state when the object is stationary and find the speed on the final section.



Show your working

KAIZEN MATHS

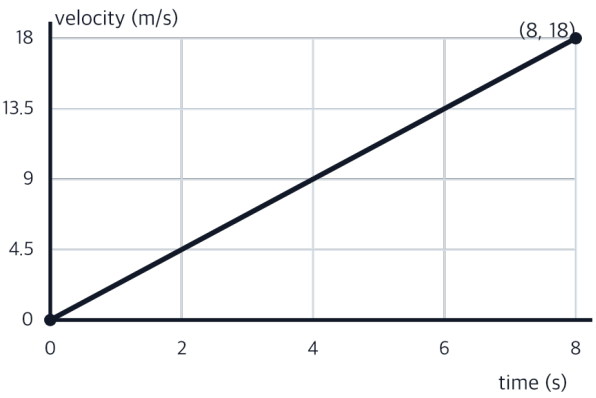
# Exit ticket

**Motion Graphs and Constant Acceleration**

Motion Graphs

Name

**EA1**




The velocity-time graph shows an object accelerating from rest. Find the distance travelled.



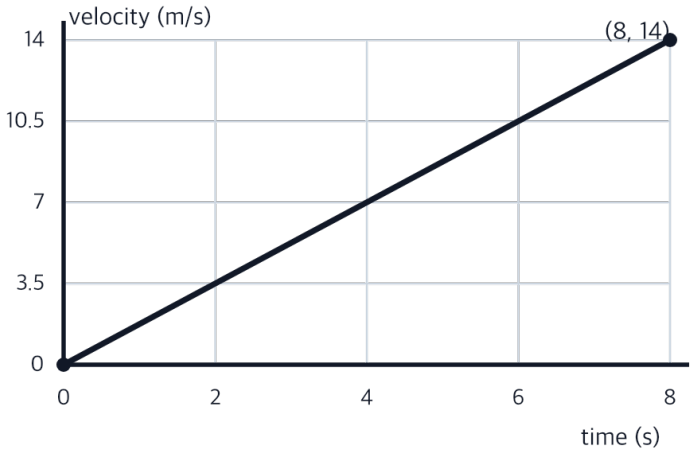
Show your working

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SA1




The velocity-time graph shows an object accelerating from rest. Find the distance travelled.

**Answer:**

56 m

**Working:**

Distance travelled is the area under the velocity-time graph.

The area under this graph is a triangle.

area = ½ × base × height

distance = ½ × 8 × 14

distance = 56

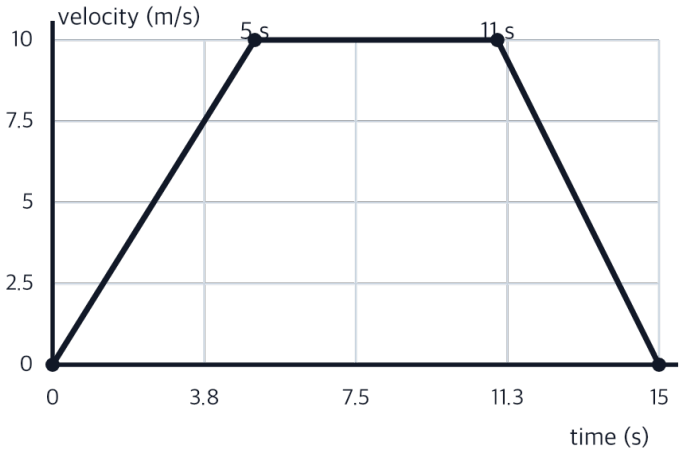
The distance travelled is 56 m.

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SB1




Use the velocity-time graph to find the total distance travelled and the average speed for the whole journey.

**Answer:**

105 m, average speed 7 m/s

**Working:**

Split the area under the graph into two triangles and one rectangle.

first triangle = ½ × 5 × 10 = 25

rectangle = (11 - 5) × 10 = 60

second triangle = ½ × (15 - 11) × 10 = 20

Add the areas to find the distance.

total distance = 25 + 60 + 20 = 105

Use average speed = total distance ÷ total time.

average speed = 105 ÷ 15 = 7

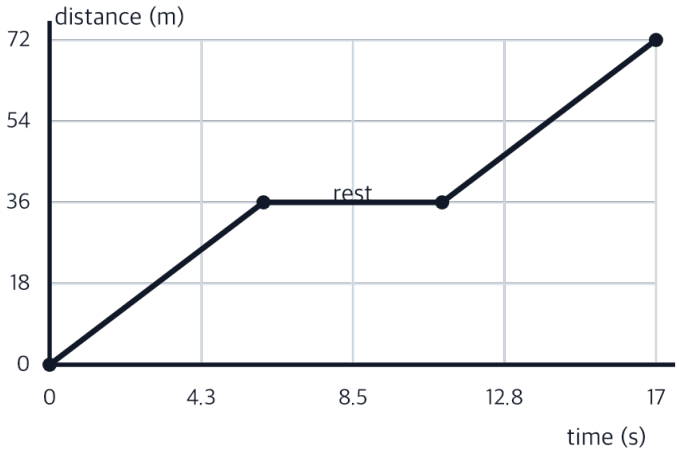
Distance = 105 m and average speed = 7 m/s.

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SC1




Use the distance-time graph to state when the object is stationary and find the speed on the final section.

**Answer:**

stationary from 6 s to 11 s; final speed 6 m/s

**Working:**

A horizontal section on a distance-time graph means the distance is not changing.

stationary interval: 6 s to 11 s

The speed on a distance-time graph is the gradient.

final section speed = change in distance ÷ change in time

speed = (72 - 36) ÷ (17 - 11)

speed = 36 ÷ 6 = 6

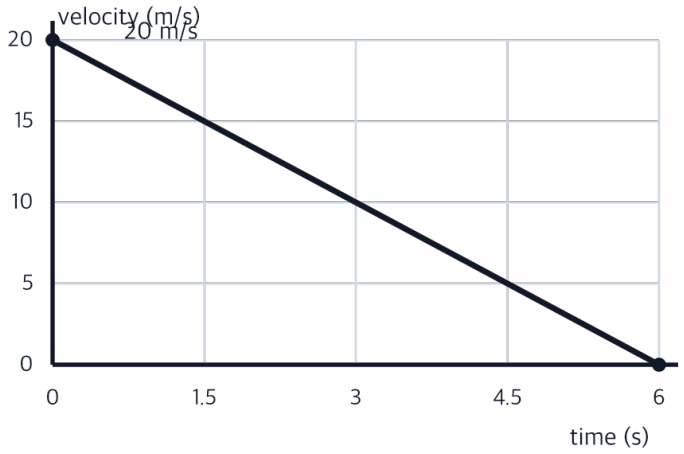
The object is stationary from 6 s to 11 s, and the final speed is 6 m/s.

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SD1




Use the velocity-time graph to find the acceleration. State the deceleration as a positive value.

**Answer:**

acceleration -3.33 m/s²; deceleration 3.33 m/s²

**Working:**

On a velocity-time graph, acceleration is the gradient.

acceleration = change in velocity ÷ change in time

$a = (0 - 20) \div 6$

$a = -3.33\ m/s^2$

Deceleration is the positive size of the negative acceleration.

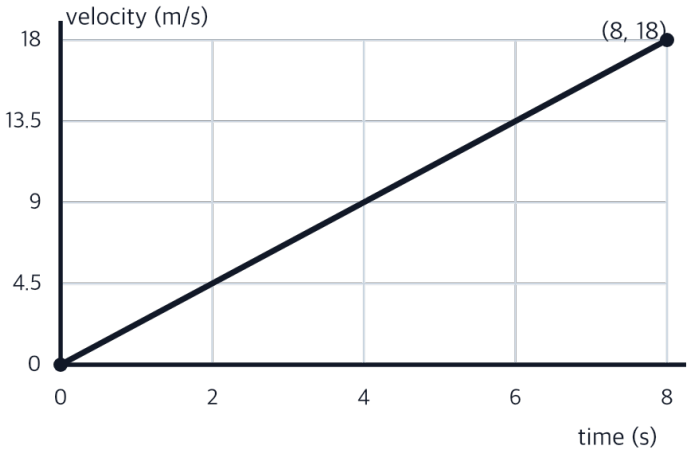
Acceleration = -3.33 m/s², so deceleration = 3.33 m/s².

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EA1




The velocity-time graph shows an object accelerating from rest. Find the distance travelled.

**Answer:**

72 m

**Working:**

Distance travelled is the area under the velocity-time graph.

The area under this graph is a triangle.

area = ½ × base × height

distance = ½ × 8 × 18

distance = 72

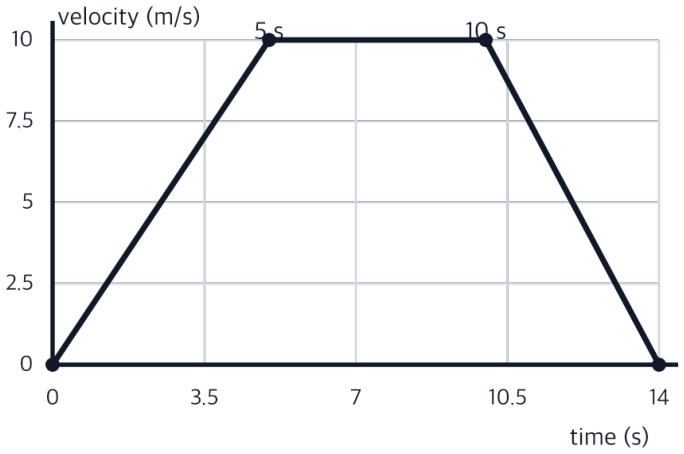
The distance travelled is 72 m.

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EB1




Use the velocity-time graph to find the total distance travelled and the average speed for the whole journey.

**Answer:**

95 m, average speed 6.79 m/s

**Working:**

Split the area under the graph into two triangles and one rectangle.

first triangle = ½ × 5 × 10 = 25

rectangle = (10 - 5) × 10 = 50

second triangle = ½ × (14 - 10) × 10 = 20

Add the areas to find the distance.

total distance = 25 + 50 + 20 = 95

Use average speed = total distance ÷ total time.

average speed = 95 ÷ 14 = 6.79

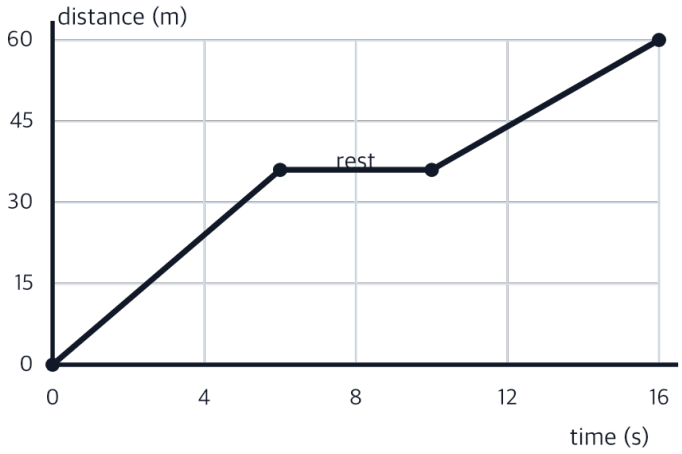
Distance = 95 m and average speed = 6.79 m/s.

# Teacher answers & working

**Motion Graphs and Constant Acceleration · Motion Graphs**

MEC-50A8-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EC1




Use the distance-time graph to state when the object is stationary and find the speed on the final section.

**Answer:**

stationary from 6 s to 10 s; final speed 4 m/s

**Working:**

A horizontal section on a distance-time graph means the distance is not changing.

stationary interval: 6 s to 10 s

The speed on a distance-time graph is the gradient.

final section speed = change in distance ÷ change in time

speed = (60 - 36) ÷ (16 - 10)

speed = 24 ÷ 6 = 4

The object is stationary from 6 s to 10 s, and the final speed is 4 m/s.